🔥 KINEMATICS – JEE ADVANCED SHORT NOTES

1. Basic Concepts

Kinematics → Motion without force

Position (x) → Location of particle

Distance → Actual path (scalar)

Displacement → Shortest path (vector)

Important: Distance ≥ |Displacement|

Displacement can be zero but distance cannot

2. Velocity & Speed

Average Velocity = Displacement / Time

Instantaneous Velocity = dx/dt

Speed = magnitude of velocity

Note: Velocity changes if direction changes

3. Acceleration

Acceleration = dv/dt

Types: Positive → speed increasing

Negative → speed decreasing

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Zero → constant velocity

4. Equations of Motion (Uniform Acceleration)

$v = u + at$

$s = ut + (1/2)at^2$

$v^2 = u^2 + 2as$

Use only when acceleration is constant

5. Graphs

Position-Time Graph: Slope = Velocity

Velocity-Time Graph: Slope = Acceleration

Area = Displacement

Acceleration-Time Graph: Area = Change in velocity

6. Relative Motion

Velocity of A w.r.t B:

$v_{AB} = v_A - v_B$

Used in: Rain problems

Boat & river problems

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7. Projectile Motion

Horizontal: Velocity constant

Vertical: Acceleration = g

Time of flight: $T = (2u \sin\theta) / g$

Maximum height: $H = (u^2 \sin^2\theta) / (2g)$

Range: $R = (u^2 \sin 2\theta) / g$

Important: Maximum range at 45°

Same range for θ and (90° − θ)

8. Vertical Motion

Take upward positive

Acceleration: $a = -g$

At highest point: $v = 0$

9. Important Tricks

Time up = Time down

Speed same at same height

Horizontal and vertical motion independent

Velocity zero ≠ acceleration zero

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10. Common Mistakes

Do not use formulas in variable acceleration

Do not mix distance and displacement

Always check sign convention

11. Advanced Topics

Relative acceleration

Projectile on inclined plane

River crossing problems

Variable acceleration (integration based)

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